

The Vigenère cipher (due to G. B. Bellaso 1553, 15 misattributed to B. de Vigenère)

Idea: Combine several shift ciphers that use different shift values

In detail:

- The plaintext and Key are assumed to be strings over the same alphabet.
- If the Key is shorter than the plaintext, cyclically repeat the characters of the Key until both strings have the same length.
- Interpret each character of the Key string as a shift parameter based on its position within the alphabet. Use this value to encrypt the corresponding character of the plaintext string.

Ex. Alphabet: A...Z Key: BAD

Message	G	R	E	E	T	I	N	G	S
Key	B	A	D	B	A	D	B	A	D
Shift	1	0	3	1	0	3	1	0	3
Ciphertext	H	R	H	F	T	L	O	G	V

Note: This was deemed "unbreakable" well into the 19th century!

Nowadays, when the key is "substantially shorter" than the message, the Vigenère cipher can be attacked/broken using frequency analysis. 16

On the other hand, we'll later see that Vigenère ciphers are provably secure under the following highly restrictive

assumptions: The key string is randomly generated, never reused, and of the same length as the plaintext message.

(What do you think of those assumptions?)

Remark: The Vigenère cipher is an early example of a "polyalphabetic substitution cipher".

A more sophisticated example is the "Enigma machine".

It was famously broken (→ "The Bombe", A. Turing et al.) and thus played a decisive role in WW II.

Reminder: divisibility, congruences, and modular arithmetic

In the following, all variables refer to integers.

Defn We say that a divides b (written: $a \mid b$) if there exists c with $ac = b$. We then also call a a divisor of b .

Ex: • $2 \mid 6$, $-3 \mid 6$, $4 \nmid 6$, $-7 \nmid 6$

- ± 1 divides every integer
- Every integer divides zero.

Defn (Gauss 1801) Two integers a, b are congruent modulo n ,

written $a \equiv b \pmod{n}$,

if $n \mid (a-b)$.

(It would be difficult to overestimate the importance of this notion!)

Ex • $\dots, -5, -1, 3, 7, 11, \dots$ are all congruent modulo 4

- $1 \equiv -1 \pmod{2}$ but $1 \not\equiv -1 \pmod{3}$

Thm Congruence modulo n is an equivalence relation on $\mathbb{Z}$ which is compatible with addition and multiplication.

- In detail:
- $a \equiv a \pmod{n}$
 - $a \equiv b \pmod{n} \implies b \equiv a \pmod{n}$
 - $a \equiv b \pmod{n}$ & $b \equiv c \pmod{n} \implies a \equiv c \pmod{n}$
 - $a \equiv a' \pmod{n}$ & $b \equiv b' \pmod{n} \implies a \pm b \equiv a' \pm b' \pmod{n}$
& $ab \equiv a'b' \pmod{n}$.

Euclidean division: let a be arbitrary and $n > 0$.

Then there are unique (!) integers q, r with $a = qn + r$ and $0 \leq r < n$. Notation: $r = a \bmod n$

Ex • $a = 72, n = 7$

Since $72 = 10 \cdot 7 + 2$, we have $72 \bmod 7 = 2$

• $-72 = (-11) \cdot 7 + 5$ so $(-72) \bmod 7 = 5$

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Defn The ring of integers modulo n consists of the set $\mathbb{Z}_n := \{0, 1, \dots, n-1\}$ and the operations $+_n$ and $\cdot_n$ defined via

$$a +_n b := \underbrace{(a + b)}_{\text{ordinary addition of integers}} \bmod n$$

$$a \cdot_n b := \underbrace{(a \cdot b)}_{\text{ordinary multiplication}} \bmod n$$

for $a, b \in \mathbb{Z}_n$.

Ex: $2 \cdot_5 3 = 1$ in $\mathbb{Z}_5$, $2 \cdot_6 3 = 0$ in $\mathbb{Z}_6$

Remarks

- When confusion is unlikely, we'll often drop the subscripts for addition and multiplication in $\mathbb{Z}_n$.
- For $a, b \in \mathbb{Z}_n$, we have $a \equiv b \pmod{n}$ iff $a = b$.
- $(\mathbb{Z}_n, +_n, \cdot_n)$ is a "ring" in the sense of abstract algebra. Informally, this means that addition & multiplication obey the "usual rules" in $\mathbb{Z}_n$. Division is a different story though...